

a Mumbai-based company. The ventilators are identical to the ones produced by the Noccarc team, with additional Avi Healthcare branding. The authors say, “as we’ve seen with the Noccarc-AVI Healthcare partnership, the start-up-SME partnership is almost like a match made in heaven. The start-up can license technology to a manufacturing SME who is then happy to utilise its idle capacity. Accelerators, and especially incubators, have the right networks and skill sets to create events and programmes where start-ups and SMEs can be brought together for match-making. SMEs’ want of innovative technology puts the promising start-ups in a sweet spot where they can benefit from these networks. This relation-



Noccarc Robotics co-founders Harshit Rathore & Nikhil Kurele
with Noccarc V310 & Noccarc H210

ship must be facilitated by incubators and accelerators as they operate with experienced manpower and systems built to further the goals of start-ups. Moreover, these organisations can help start-ups and SMEs form an agreement and resolve issues if any. Such associations can also help the start-up stay asset-light and focus on technological innovation.”

One of the things that did not work out was partnership with a PSU, which tried to strong arm the fresh startup into giving it an exclusive license for manufacturing the ventilator. The deal ultimately did not go through. A startup partnering with a PSU may be a difficult undertaking, but can actually benefit both the parties involved. The authors explain, “This is harder than start-ups partnering with a SME because

PSUs are behemoths whose slow pace can hugely frustrate a start-up. But, we see many PSUs, especially the Maharatnas, becoming very conscious of being innovative and relevant in a fast-changing world. For a PSU partnership to work well, the top management at the PSU has to be fully supportive and invested. Besides this, a start-up needs to be sure of why it is partnering with a PSU. Is it because of its manufacturing capacity (e.g. for ventilators), for accessing its distribution channels (e.g. several oil PSUs have tied up or invested in start-ups), or is it for some other reason.”

Another one of the interesting ideas presented in the book is that startups might just be better off going for value addition as against 100

percent indigeneness. This means that by deciding which features to use, targeting a particular quality and price point, and then importing the necessary components to manufacture a product that can be sold in markets around the world is something startups can focus on instead of going for making

the product entirely from scratch using locally sourced components only. The authors explain, “there is no point in trying to make every single component or raw material in India especially where the domestic volume requirements are not very large. It’s preferable to allow commodities, such as raw materials and components, come in from other countries. Indian companies can focus on designing and manufacturing the actual end product, and infuse it with software and other IP that makes the finished product truly valuable. This will ensure that most of the profits, and all the IP is retained in India.”

We then asked about who can benefit from the new model for rapid innovation introduced in the book, “The agile innovation model and the ‘Silicon Valley Reinvented’ model

is very relevant for start-ups, SMEs and large corporations. It shows how an organisation can operate under a hugely transformative paradigm of innovation and execute it at record speed. It shows how super ambitious goals can be set, how a diverse location-agnostic task force can be brought together, and how executional rigour can be achieved. For organisations in India, it’s a chance to set the pace for the rest of the world with a whole-new innovation paradigm.”

Skanray Technologies, one of the existing medical equipment manufacturers in India, and also mentioned in the book, boosted the production of its ICU ventilators. The Indian Institute of Science developed its own ICU ventilator with a custom pneumatic system, called Praana and made the schematics publicly available by 1 April 2020. The ventilator went from a drawing board to a proof of concept system within 35 days, and two additional weeks to build the working prototype. An IIT Roorkee Team developed the low cost “Prana-Vayu” ventilator which has a circuit breaker that opens the valve to ambient air if there was too much pressure build up. IIT Bhubaneshwar came up with a volume control mode ventilator, known as Patient Responsive Active Assist coNtrol (PRAAN).

Zulqarnain, an IIT Bombay student from Srinagar who had gone back home during the pandemic realised that there were less than a hundred ventilators in the valley, and teamed up with friends from NIT Srinagar and Islamic University of Science & Technology (IUST), developed a low cost ventilator known as Ruhdaar within a month of the lockdown, using laser cutting, 3D printing and sensors available in house at IUST for the prototype. The ventilator was another unique design which used mechanical compression of a bulb. Zulqarnain says, “Ruhdaar provides necessary functionalities which can provide adequate breathing support necessary to save the life of a critically ill COVID-19 patient.” Engineers from IIT Mandi developed a low cost portable ventilator that combined a life saving AMBU bag with mecha-